Corporate Finance — Problem Set 2

Worked solutions

Statement: `Listas/PS2.pdf`. Each exercise is restated briefly, then solved step by step. Rates are written as decimals inside equations. References are to the lecture slides (`Aula/Topic2A.pdf`, `Aula/Topic2B.pdf`) and to Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011). Every number below is reproduced by `Listas/PS2-solutions_check.py`, which also generates the table of Appendix A. A short answer key is in `Listas/PS2-solutions-short.pdf`. The statement labels two parts of Exercise 1 as "(e)"; the second is called (e′) here.

1. Exercise 1 — State prices, options and MM in a complete market

Data. Two equally likely states at $t = 1$, slump (s) and boom (b). Stock: price 1.05, payoff $(0,\ 2.5)$. Bond: price 0.92, payoff $(1,\ 1)$. Project: investment 1 at $t = 0$, payoff $(0.5,\ 3)$. MM assumptions hold, no taxes. The entrepreneur has no money.

Notation.

- $\psi_s, \psi_b > 0$: state prices, the $t = 0$ price of 1 paid only in the slump (boom).
- r_f: risk-free rate; q_s, q_b: risk-neutral probabilities, $q_\omega = \psi_\omega(1 + r_f)$.
- $X = (X_s, X_b) = (0.5,\ 3)$: project payoff; A: market value of the project (the firm's assets).
- $C(K)$, $P(K)$: prices of a call and a put on the project with strike $K \geq 0$.
- D: amount raised with debt (= market value of debt); F: face value of debt.
- E: market value of equity; r_A, r_D, r_E: expected returns, computed with the true probabilities $\frac{1}{2}, \frac{1}{2}$.

(a) State prices

Step 1 — Pricing equations. A traded asset's price is the sum of its payoffs times state prices:

$$\text{Stock:}\quad 0\,\psi_s + 2.5\,\psi_b = 1.05, \tag{1.1}$$

$$\text{Bond:}\quad 1\,\psi_s + 1\,\psi_b = 0.92. \tag{1.2}$$

Step 2 — Solve. From (1.1):

$$\psi_b = \frac{1.05}{2.5} = 0.42. \tag{1.3}$$

Substituting (1.3) into (1.2):

$$\psi_s + (0.42) = 0.92 \quad\Longrightarrow\quad \psi_s = 0.50. \tag{1.4}$$

Step 3 — Implied rates and probabilities. The bond pays 1 for sure, so $1 + r_f = 1/0.92$, i.e. $r_f = 0.086957$ (8.70%). The risk-neutral probabilities are $q_s = 0.50/0.92 = 0.543478$ and $q_b = 0.42/0.92 = 0.456522$.

Answer. The slump claim costs $\psi_s = 0.50$ and the boom claim $\psi_b = 0.42$.

Intuition. The two states are equally likely, yet a unit in the slump costs more than a unit in the boom ($0.50 > 0.42$). Investors value consumption most when they are poor; insurance against the slump is expensive. Equivalently, the risk-neutral probability of the slump (54.3%) exceeds its true probability (50%).

Step 4 — Value of the project. Using (1.3)–(1.4):

$$\begin{aligned} A &= (0.5)(0.50) + (3)(0.42) \\ &= 0.25 + 1.26 = 1.51. \end{aligned} \tag{1.5}$$

The project costs 1 and is worth 1.51: NPV = 0.51.

(b) Calls on the project

A call with strike K pays $\max(X_\omega - K, 0)$ in state ω.

$$C(1) = \max(0.5-1,0)(0.50) + \max(3-1,0)(0.42) = 0 + (2)(0.42) = 0.84, \tag{1.6}$$
$$C(1.5) = \max(0.5-1.5,0)(0.50) + \max(3-1.5,0)(0.42) = 0 + (1.5)(0.42) = 0.63. \tag{1.7}$$

(c) Puts on the project

A put with strike K pays $\max(K - X_\omega, 0)$.

$$P(1) = \max(1-0.5,0)(0.50) + \max(1-3,0)(0.42) = (0.5)(0.50) + 0 = 0.25, \tag{1.8}$$
$$P(1.5) = \max(1.5-0.5,0)(0.50) + \max(1.5-3,0)(0.42) = (1)(0.50) + 0 = 0.50. \tag{1.9}$$

(d) Put–call parity and MM

Step 1 — The middle term. $K/(1/0.92) = 0.92K$ is the price of a safe bond paying K at $t = 1$, since $1/0.92 = 1 + r_f$.

Step 2 — Substitute (1.6)–(1.9)**.**

$$C(1) + 0.92(1) - P(1) = (0.84) + 0.92 - (0.25) = 1.51, \tag{1.10}$$
$$C(1.5) + 0.92(1.5) - P(1.5) = (0.63) + 1.38 - (0.50) = 1.51. \tag{1.11}$$

Both sides equal A from (1.5).

Step 3 — Why it holds for every K. State by state, for any payoff X_ω and strike K:

$$\max(X_\omega - K, 0) + K - \max(K - X_\omega, 0) = X_\omega. \tag{1.12}$$

(If $X_\omega \geq K$: $(X_\omega - K) + K - 0$; if $X_\omega < K$: $0 + K - (K - X_\omega)$.) Pricing both sides with state prices gives $C(K) + K/(1 + r_f) - P(K) = A$ for every K: put–call parity.

Answer. Both sides equal $1.51 = A$, the value of the project, for any strike.

Intuition. Read K as the face value of debt. With limited liability, equity pays $\max(X - K, 0)$: it is a call on the assets, worth $C(K)$. Risky debt pays $\min(X, K) = K - \max(K - X, 0)$: a safe bond minus a put, worth $K/(1 + r_f) - P(K)$. The put is the creditors' loss from limited liability. By (1.12) equity plus debt is worth A whatever K is. A higher face value moves value from the call to the bond, but the total does not change. This is MM I, proved by put–call parity.

Ref: Slides Topic 2A (complete markets, state prices, MM I with risky debt); Berk §21.5–21.6 (put–call parity; equity as a call), §16.1–16.2.

(e) All debt: raise 1 with debt

Step 1 — Is the debt safe?. If $F \leq 0.5$ the debt pays F in both states and is worth $0.92F$. Raising 1 would need $F = 1/0.92 = 1.087 > 0.5$, a contradiction. So the debt defaults in the slump: it pays $\min(X, F) = (0.5,\ F)$ with $0.5 < F \leq 3$.

Step 2 — Face value. Debt must be worth what it raises:

$$(0.5)(0.50) + F(0.42) = 1 \quad \Longrightarrow \quad F = \frac{1 - 0.25}{0.42} = 1.785714. \tag{1.13}$$

Step 3 — Values. The firm's assets are the project, $A = 1.51$ (1.5); $D = 1$; and

$$E = A - D = 1.51 - 1 = 0.51 = C(F) = (3 - 1.785714)(0.42). \tag{1.14}$$

Step 4 — Expected returns. Expected payoffs use the true probabilities $\frac{1}{2}$:

$$r_A = \frac{0.5(0.5) + 0.5(3)}{1.51} - 1 = \frac{1.75}{1.51} - 1 = 0.158940, \tag{1.15}$$

$$r_D = \frac{0.5(0.5) + 0.5(1.785714)}{1} - 1 = 1.142857 - 1 = 0.142857, \tag{1.16}$$

$$r_E = \frac{0.5(0) + 0.5(3 - 1.785714)}{0.51} - 1 = \frac{0.607143}{0.51} - 1 = 0.190476. \tag{1.17}$$

Step 5 — Check MM. The value-weighted average of (1.16) and (1.17) must equal (1.15): $\frac{1}{1.51}(0.142857) + \frac{0.51}{1.51}(0.190476) = 0.094607 + 0.064333 = 0.158940$. ✓

Answer. $F = 1.7857$; $A = 1.51$, $D = 1$, $E = 0.51$; $r_A = 15.89\%$, $r_D = 14.29\%$, $r_E = 19.05\%$.

Remark. The promised yield, $F/D - 1 = 78.57\%$, is far above the expected return on debt, 14.29%: creditors are promised 1.786 but receive only 0.5 in the slump. The yield is not the cost of debt. The entrepreneur keeps the whole equity, worth 0.51: exactly the project's NPV.

(e′) Any mix: raise D with debt and $1 - D$ with equity

Step 1 — Face value as a function of D. Debt is safe as long as $F \le 0.5$, i.e. $D = 0.92F \le 0.46$.

$$F(D) = \begin{cases} \dfrac{D}{0.92}, & 0 < D \le 0.46 \quad \text{(safe debt)}, \\ \dfrac{D - (0.5)(0.50)}{0.42} = \dfrac{D - 0.25}{0.42}, & 0.46 < D \le 1.51 \quad \text{(risky debt)}. \end{cases} \tag{1.18}$$

The two branches meet at $D = 0.46$, where both give $F = 0.5$.

Step 2 — Values. $A = 1.51$ for every D, and $E = 1.51 - D$. Outside shareholders pay $1 - D$ for their stake, so the entrepreneur's stake is worth $(1.51 - D) - (1 - D) = 0.51$, the NPV, for every mix.

Step 3 — Returns, safe region ($D \le 0.46$). $r_D = r_f = 8.70\%$ and $r_A = 15.89\%$ (1.15). MM II follows from $A\, r_A = D\, r_D + E\, r_E$:

$$r_E = r_A + \frac{D}{E}\,(r_A - r_D). \tag{1.19}$$

For example, $D = 0.20$: $F = 0.20/0.92 = 0.217391$, $E = 1.31$, and by (1.19)

$$r_E = 0.158940 + \frac{0.20}{1.31}(0.071984) = 0.158940 + (0.152672)(0.071984) = 0.169930.$$

Step 4 — Returns, risky region ($D > 0.46$). Debt pays $(0.5,\ F)$, so

$$r_D = \frac{0.5(0.5) + 0.5\,F(D)}{D} - 1, \tag{1.20}$$

rising with D (8.92% at $D = 0.47$, 14.29% at $D = 1$). Equity pays $(0,\ 3 - F)$: nothing in the slump. It is a multiple of the pure boom claim, so its expected return is that of the boom claim,

$$r_E = \frac{0.5}{\psi_b} - 1 = \frac{0.5}{0.42} - 1 = 0.190476, \tag{1.21}$$

the same for every D in this region. Equation (1.19) still holds, now with the rising r_D.

Answer. $F(D)$ is given by (1.18); $A = 1.51$ and $r_A = 15.89\%$ for every D; $E = 1.51 - D$. For $D \le 0.46$: $r_D = 8.70\%$ and r_E rises linearly in D/E from 15.89% to 19.05%. For $D > 0.46$: r_D rises from 8.70% to 14.29% and r_E stays at 19.05%. The full table for $D = 0.01, \ldots, 1.00$ is in Appendix A.

Intuition. This is MM II with risky debt. While debt is safe, each extra unit of leverage loads all the asset risk onto equity, so r_E rises linearly. Once debt defaults in the slump, creditors take the slump payoff and start bearing risk. Equity is left with a boom-only claim whose risk can grow no further, so r_E flattens and the extra leverage shows up in a rising r_D. Through it all, the WACC equals r_A and the firm is worth 1.51.

(f) Debt-financed dividend

Step 1 — Face value. Raising 1.1 is in the risky region of (1.18): $F = (1.1 - 0.25)/0.42 = 2.023810$.

Step 2 — Value of the firm. After the dividend, the firm holds only the project: $A = 1.51$, split as $D = 1.1$ and $E = 1.51 - 1.1 = 0.41$. (Before the dividend it also held 0.1 of cash, so it was worth 1.61.)

Step 3 — The entrepreneur's wealth. Dividend plus equity: $0.1 + 0.41 = 0.51$, exactly what she had with no dividend (1.14).

Answer. The firm is worth 1.51 after the dividend ($D = 1.1$, $E = 0.41$). The entrepreneur is not better off: her wealth stays at 0.51. The dividend only moves 0.1 of it from $t = 1$ to $t = 0$.

Intuition. The creditors pay a fair 1.1 for a claim worth 1.1, so they transfer no value. The cash paid out is the entrepreneur's own money, taken out early and paid for with a higher face value. She could get the same $t = 0$ consumption by selling 0.1 of her equity. With complete markets and no frictions, payout and capital structure only change the timing of her wealth, not its amount. This is the dividend-irrelevance argument of Topic 2B.

2. Exercise 2 — Leverage, WACC and three debt policies

Data (millions of dollars). 10 million shares, no debt; equity beta 1.20; FCF = 25 per year in perpetuity from next year. Target $D/E = 0.25$, continuously rebalanced; $r_D = 0.03$; $\tau_c = 0.25$; CAPM with $r_f = 0.03$ and $E[R_m] = 0.08$.

Notation.

- r_U, β_U: expected return and beta of the unlevered firm (its assets).
- r_E, β_E: of levered equity; $r_D = 0.03$: cost of debt; $\tau_c = 0.25$: tax rate.
- V_U, V_L: values of the unlevered and levered firm; V_{TS}: present value of the interest tax shields.
- D, E: market values of debt and equity; N: number of shares; p: share price.

(a) Cost of equity before and after

Step 1 — Before (all equity). By the CAPM:

$$r_U = r_f + \beta_U(E[R_m] - r_f) = 0.03 + 1.2(0.08 - 0.03) = 0.09. \tag{2.1}$$

Step 2 — After. With continuous rebalancing, debt moves with firm value, so the tax shields are as risky as the assets: $r_{TS} = r_U$ (slides, Topic 2B). Then the pre-tax WACC equals r_U, and MM II keeps its no-tax form:

$$\begin{aligned} r_E &= r_U + \frac{D}{E}(r_U - r_D) \\ &= 0.09 + 0.25(0.09 - 0.03) = 0.105. \end{aligned} \tag{2.2}$$

Check through betas: $r_D = r_f$ means $\beta_D = 0$, so $\beta_E = \beta_U(1 + D/E) = 1.2 \times 1.25 = 1.5$ and $r_E = 0.03 + 1.5(0.05) = 0.105$.

Answer. $r_E = 9\%$ before and 10.5% after.

(b) New WACC

$D/E = 0.25$ means $D/V = 0.25/1.25 = 0.20$ and $E/V = 0.80$. Using (2.2):

$$\begin{aligned} r_{WACC} &= \frac{E}{V} r_E + \frac{D}{V} r_D (1 - \tau_c) \\ &= (0.80)(0.105) + (0.20)(0.03)(0.75) = 0.084 + 0.0045 = 0.0885. \end{aligned} \tag{2.3}$$

Check with the slides' form for continuous rebalancing:

$$r_{WACC} = r_U - \frac{D}{V} \tau_c r_D = 0.09 - (0.20)(0.25)(0.03) = 0.0885. \quad \checkmark$$

Answer. $r_{WACC} = 8.85\%$.

(c) Values, price and repurchase

Step 1 — Before. Using (2.1):

$$V_U = \frac{25}{0.09} = 277.78, \qquad p_{\text{before}} = \frac{277.78}{10} = 27.78. \tag{2.4}$$

Step 2 — Levered firm and tax shield. Using (2.3):

$$V_L = \frac{25}{0.0885} = 282.49, \qquad V_{TS} = V_L - V_U = 282.49 - 277.78 = 4.71. \tag{2.5}$$

Step 3 — Price at the announcement. When the plan is announced, no debt has been issued yet. The shareholders own the assets plus the expected tax shields, so the whole gain goes to them at once:

$$p = \frac{V_U + V_{TS}}{N} = \frac{282.49}{10} = 28.25. \tag{2.6}$$

Step 4 — Issue and repurchase. The debt issued is $D = 0.20\, V_L = 0.20(282.49) = 56.50$. It buys back shares at the fair price (2.6):

$$\text{Shares repurchased} = \frac{56.50}{28.25} = 2.00, \qquad N_{\text{after}} = 10 - 2 = 8.00. \tag{2.7}$$

Check: $E = 0.80(282.49) = 225.99$ and $225.99/8 = 28.25$; the repurchase leaves the price unchanged.

Answer. $V_U = 277.78$; price 27.78 before, 28.25 after the announcement; $V_L = 282.49$, $V_{TS} = 4.71$; 2 million shares repurchased with \$56.50 million of debt; 8 million shares remain, at \$28.25.

(d) Interest equal to 20% of FCF

Interest is $0.20 \times 25 = 5$ per year, so the tax shield is $\tau_c \times 5 = 0.25 \times 5 = 1.25$ per year. Because interest is tied to FCF, the shield is exactly as risky as FCF and is discounted at r_U (2.1):

$$V_{TS} = \frac{1.25}{0.09} = 13.89. \tag{2.8}$$

Answer. $V_{TS} =$ \$13.89 million.

Remark. Interest of 5 at $r_D = 3\%$ corresponds to debt of about $5/0.03 = 166.67$ on average. That is far more debt than in (c), hence the larger shield. The discount rate is still r_U, not r_D, because the amount of debt moves with the firm's cash flows.

(e) $100 million of perpetual debt

Step 1 — Tax shield. The debt is fixed and riskless, so the shield $\tau_c r_D D$ is discounted at r_D:

$$V_{TS} = \frac{\tau_c r_D D}{r_D} = \tau_c D = 0.25(100) = 25. \tag{2.9}$$

Step 2 — Price and shares. $V_L = 277.78 + 25 = 302.78$. At the announcement $p = 302.78/10 = 30.28$. The repurchase buys $100/30.28 = 3.30$ million shares, leaving $10 - 3.30 = 6.70$ million. Check: $E = 302.78 - 100 = 202.78$ and $202.78/6.70 = 30.28$.

Answer. V_{TS} = \$25 million; price \$30.28; 3.30 million shares repurchased; 6.70 million remain.

Intuition. The three policies give shields of 4.71, 13.89 and 25. Two things drive the value: how much debt there is, and how risky the shield is. A fixed amount of debt makes the shield as safe as the debt; a policy tied to firm value or FCF makes it as risky as the assets. This is why the WACC formula depends on the debt policy, and why APV is the safer tool when debt follows a known schedule.

Ref: Slides Topic 2B (value of the tax shield; continuous rebalancing; WACC formulas and appendix); Berk §16.3 (the interest tax deduction and firm value; value of the interest tax shield; permanent debt; leverage and the WACC with taxes), pp. 473–478.

3. Exercise 3 — Risky debt and who pays for bankruptcy costs

Data. Investment $I = 60$; FCF = 100 or 60 with probability $\frac{1}{2}$ each; $r_A = 0.15$; no taxes or other frictions.

Notation.

- V_U, V_L: value of the all-equity and the levered firm; $D = 60$: amount borrowed.
- F: face value of debt; r_D: expected return demanded by creditors; E: value of equity.
- BC: bankruptcy cost paid in default, a fraction 0.07 of the realised cash flow.

(a) All-equity firm

$$V_U = \frac{0.5(100) + 0.5(60)}{1.15} = \frac{80}{1.15} = 69.57. \tag{3.1}$$

The project's NPV is $69.57 - 60 = 9.57$.

(b) Debt of 60 at an expected return of 6.6%

Step 1 — Expected payoff creditors need. $60(1.066) = 63.96$.

Step 2 — Face value. If the debt were safe, $F = 63.96 > 60$, so it could not be paid in the bad state: the debt is risky. Creditors get 60 (everything) in the bad state and F in the good one:

$$0.5(60) + 0.5F = 63.96 \quad \Longrightarrow \quad F = 2(63.96) - 60 = 67.92. \tag{3.2}$$

Step 3 — Equity. By MM the levered firm is worth V_U, so

$$E = V_L - D = 69.57 - 60 = 9.57. \tag{3.3}$$

Equity pays $100 - 67.92 = 32.08$ in the good state and 0 in the bad state. Its expected payoff is 16.04, so $r_E = 16.04/9.57 - 1 = 67.69\%$. Check: $\frac{60}{69.57}(0.066) + \frac{9.57}{69.57}(0.6769) = 0.15 = r_A$.

Answer. $F = 67.92$; $E = 9.57$ (the NPV, kept by the entrepreneur).

(c) Bankruptcy costs of 7% of the realised cash flow

Step 1 — What creditors get in default. Default happens in the bad state ($60 < F$). Lawyers take $0.07(60) = 4.2$, so creditors receive 55.8.

Step 2 — Face value. Keeping the creditors' required expected return at 6.6%:

$$0.5(55.8) + 0.5F = 63.96 \quad \Longrightarrow \quad F = 2(63.96) - 55.8 = 72.12. \tag{3.4}$$

Step 3 — Value of the firm and of equity. Bankruptcy costs are a deadweight loss. The expected cost is $0.5(4.2) = 2.1$. It is a part of the project's cash flow, so we discount it at r_A:

$$V_L = V_U - PV(BC) = \frac{80 - 2.1}{1.15} = 69.57 - 1.83 = 67.74, \tag{3.5}$$

$$E = V_L - D = 67.74 - 60 = 7.74. \tag{3.6}$$

Step 4 — Who pays. Creditors still get a fair expected return on their 60; the higher face value (3.4) compensates them for the 4.2 lost in default. Equity falls from 9.57 (3.3) to 7.74 (3.6), a loss of $1.83 = PV(BC)$.

Answer. $F = 72.12$; $E = 7.74$. The shareholders (the entrepreneur) pay the whole present value of the bankruptcy costs, 1.83. They pay it ex ante, through the higher face value, even though the lawyers are paid out of the creditors' recovery in default.

Intuition. Creditors anticipate the loss in default and price it in. The cost therefore falls on whoever raises the money, at the time it is raised. This is why expected distress costs enter the firm's capital-structure choice (the trade-off theory), even though shareholders get nothing in default.

Remark. A fully consistent pricing gives the same answer to the question "who pays". The market data of (a) and (b) imply state prices $\psi_{good} = 0.2982$ and $\psi_{bad} = 0.6625$ (risk-free rate 4.10%). The bad-state cost is then worth $4.2 \times 0.6625 = 2.78$, and equity is worth $69.57 - 2.78 - 60 = 6.78$. Debt worth 60 would need $F = 77.25$, so the creditors' expected return rises to 10.88% because their claim is now riskier. The main answer keeps 6.6% and discounts at r_A, as the statement suggests. Either way, shareholders bear the full present value of the cost.

Ref: Slides Topic 2B (bankruptcy costs; $V_L = V_U + V_{TS} - PV$(bankruptcy costs)); Berk §16.4–16.5 (costs of bankruptcy and financial distress; the tradeoff theory), pp. 479–482.

4. Exercise 4 — Yerba Industries: leverage, EPS and the P/E

Data. All-equity, $\beta_U = 1.2$, $r_U = 0.125$. Issues risk-free debt at 0.05 and buys back 40% of its shares. Perfect markets. Before: expected EPS £1.50, forward P/E 14.

Notation.

- $D/V = 0.40$, $E/V = 0.60$: weights after the transaction (the repurchase swaps 40% of the equity for debt at market value, so V does not change); $D/E = 0.4/0.6 = 2/3$.
- $\beta_D = 0$, $r_D = 0.05$: risk-free debt; β_E, r_E: of equity after the transaction.

(a) Equity beta after

With $\beta_D = 0$, the asset beta is $\beta_U = \frac{E}{V}\beta_E$, so

$$\beta_E = \beta_U \left(1 + \frac{D}{E}\right) = 1.2 \left(1 + \frac{0.4}{0.6}\right) = 2.0. \tag{4.1}$$

(b) Expected return on equity after

By MM II:

$$r_E = r_U + \frac{D}{E}(r_U - r_D) = 0.125 + \frac{2}{3}(0.125 - 0.05) = 0.175. \tag{4.2}$$

Check with the CAPM: the market risk premium is $(0.125 - 0.05)/1.2 = 0.0625$, so $r_E = 0.05 + 2.0(0.0625) = 0.175$.

(c) Earnings per share after

Step 1 — Price. Before: $p = 1.50 \times 14 = £21$. Under MM the transaction does not change it.

Step 2 — Per original share. Debt raised per original share: $0.40 \times 21 = 8.40$. Interest: $0.05 \times 8.40 = 0.42$. Earnings left for equity: $1.50 - 0.42 = 1.08$, now shared by 0.60 shares:

$$EPS_{\text{after}} = \frac{1.08}{0.60} = £1.80. \tag{4.3}$$

Answer. EPS rises from £1.50 to £1.80 (+20%), but shareholders do not benefit: the share price stays at £21. The higher expected EPS is exactly the compensation for the higher risk of leveraged equity (β_E from 1.2 to 2.0, r_E from 12.5% to 17.5%).

(d) P/E after

$$P/E_{\text{after}} = \frac{21}{1.80} = 11.67. \tag{4.4}$$

Answer. The forward P/E falls from 14 to 11.67. This is reasonable: each pound of expected earnings is now riskier, so investors pay less for it. The earnings yield rises from $1.50/21 = 7.14\%$ to $1.80/21 = 8.57\%$, in line with the higher cost of equity.

Intuition. Leverage raises EPS whenever the return on assets exceeds the cost of debt, and that says nothing about value. Price is expected earnings capitalised at a rate that reflects their risk; MM II says the rate rises just enough to offset the higher EPS. A manager who targets EPS or ROE growth through leverage creates no value in a perfect market (the slides' "borrow more, value rises" fallacy).

Ref: Slides Topic 2A (MM II; leverage and EPS); Berk §16.2 (the effect of leverage on risk and return; leverage and the cost of capital; capital structure fallacies), pp. 467–471.

A. Exercise 1(e′): full table for $D = 0.01, \ldots, 1.00$

Generated by `Listas/PS2-solutions_check.py`, which also checks $\frac{D}{A}r_D + \frac{E}{A}r_E = r_A$ on every row. Debt is safe up to $D = 0.46$ ($F \leq 0.5$) and risky beyond. Returns in %.

D	F	r_D	r_E	r_A	E	A
0.01	0.0109	8.70	15.94	15.89	1.50	1.51
0.02	0.0217	8.70	15.99	15.89	1.49	1.51
0.03	0.0326	8.70	16.04	15.89	1.48	1.51
0.04	0.0435	8.70	16.09	15.89	1.47	1.51
0.05	0.0543	8.70	16.14	15.89	1.46	1.51
0.06	0.0652	8.70	16.19	15.89	1.45	1.51
0.07	0.0761	8.70	16.24	15.89	1.44	1.51
0.08	0.0870	8.70	16.30	15.89	1.43	1.51
0.09	0.0978	8.70	16.35	15.89	1.42	1.51
0.10	0.1087	8.70	16.40	15.89	1.41	1.51
0.11	0.1196	8.70	16.46	15.89	1.40	1.51

D	F	r_D	r_E	r_A	E	A
0.12	0.1304	8.70	16.52	15.89	1.39	1.51
0.13	0.1413	8.70	16.57	15.89	1.38	1.51
0.14	0.1522	8.70	16.63	15.89	1.37	1.51
0.15	0.1630	8.70	16.69	15.89	1.36	1.51
0.16	0.1739	8.70	16.75	15.89	1.35	1.51
0.17	0.1848	8.70	16.81	15.89	1.34	1.51
0.18	0.1957	8.70	16.87	15.89	1.33	1.51
0.19	0.2065	8.70	16.93	15.89	1.32	1.51
0.20	0.2174	8.70	16.99	15.89	1.31	1.51
0.21	0.2283	8.70	17.06	15.89	1.30	1.51
0.22	0.2391	8.70	17.12	15.89	1.29	1.51
0.23	0.2500	8.70	17.19	15.89	1.28	1.51
0.24	0.2609	8.70	17.25	15.89	1.27	1.51
0.25	0.2717	8.70	17.32	15.89	1.26	1.51
0.26	0.2826	8.70	17.39	15.89	1.25	1.51
0.27	0.2935	8.70	17.46	15.89	1.24	1.51
0.28	0.3043	8.70	17.53	15.89	1.23	1.51
0.29	0.3152	8.70	17.61	15.89	1.22	1.51
0.30	0.3261	8.70	17.68	15.89	1.21	1.51
0.31	0.3370	8.70	17.75	15.89	1.20	1.51
0.32	0.3478	8.70	17.83	15.89	1.19	1.51
0.33	0.3587	8.70	17.91	15.89	1.18	1.51
0.34	0.3696	8.70	17.99	15.89	1.17	1.51
0.35	0.3804	8.70	18.07	15.89	1.16	1.51
0.36	0.3913	8.70	18.15	15.89	1.15	1.51
0.37	0.4022	8.70	18.23	15.89	1.14	1.51
0.38	0.4130	8.70	18.31	15.89	1.13	1.51
0.39	0.4239	8.70	18.40	15.89	1.12	1.51
0.40	0.4348	8.70	18.49	15.89	1.11	1.51
0.41	0.4457	8.70	18.58	15.89	1.10	1.51
0.42	0.4565	8.70	18.67	15.89	1.09	1.51
0.43	0.4674	8.70	18.76	15.89	1.08	1.51
0.44	0.4783	8.70	18.85	15.89	1.07	1.51
0.45	0.4891	8.70	18.95	15.89	1.06	1.51
0.46	0.5000	8.70	19.05	15.89	1.05	1.51
0.47	0.5238	8.92	19.05	15.89	1.04	1.51
0.48	0.5476	9.13	19.05	15.89	1.03	1.51
0.49	0.5714	9.33	19.05	15.89	1.02	1.51
0.50	0.5952	9.52	19.05	15.89	1.01	1.51
0.51	0.6190	9.71	19.05	15.89	1.00	1.51
0.52	0.6429	9.89	19.05	15.89	0.99	1.51
0.53	0.6667	10.06	19.05	15.89	0.98	1.51
0.54	0.6905	10.23	19.05	15.89	0.97	1.51
0.55	0.7143	10.39	19.05	15.89	0.96	1.51
0.56	0.7381	10.54	19.05	15.89	0.95	1.51
0.57	0.7619	10.69	19.05	15.89	0.94	1.51
0.58	0.7857	10.84	19.05	15.89	0.93	1.51
0.59	0.8095	10.98	19.05	15.89	0.92	1.51
0.60	0.8333	11.11	19.05	15.89	0.91	1.51
0.61	0.8571	11.24	19.05	15.89	0.90	1.51
0.62	0.8810	11.37	19.05	15.89	0.89	1.51
0.63	0.9048	11.49	19.05	15.89	0.88	1.51
0.64	0.9286	11.61	19.05	15.89	0.87	1.51
0.65	0.9524	11.72	19.05	15.89	0.86	1.51
0.66	0.9762	11.83	19.05	15.89	0.85	1.51
0.67	1.0000	11.94	19.05	15.89	0.84	1.51
0.68	1.0238	12.04	19.05	15.89	0.83	1.51
0.69	1.0476	12.15	19.05	15.89	0.82	1.51
0.70	1.0714	12.24	19.05	15.89	0.81	1.51
0.71	1.0952	12.34	19.05	15.89	0.80	1.51
0.72	1.1190	12.43	19.05	15.89	0.79	1.51
0.73	1.1429	12.52	19.05	15.89	0.78	1.51
0.74	1.1667	12.61	19.05	15.89	0.77	1.51

D	F	r_D	r_E	r_A	E	A
0.75	1.1905	12.70	19.05	15.89	0.76	1.51
0.76	1.2143	12.78	19.05	15.89	0.75	1.51
0.77	1.2381	12.86	19.05	15.89	0.74	1.51
0.78	1.2619	12.94	19.05	15.89	0.73	1.51
0.79	1.2857	13.02	19.05	15.89	0.72	1.51
0.80	1.3095	13.10	19.05	15.89	0.71	1.51
0.81	1.3333	13.17	19.05	15.89	0.70	1.51
0.82	1.3571	13.24	19.05	15.89	0.69	1.51
0.83	1.3810	13.31	19.05	15.89	0.68	1.51
0.84	1.4048	13.38	19.05	15.89	0.67	1.51
0.85	1.4286	13.45	19.05	15.89	0.66	1.51
0.86	1.4524	13.51	19.05	15.89	0.65	1.51
0.87	1.4762	13.57	19.05	15.89	0.64	1.51
0.88	1.5000	13.64	19.05	15.89	0.63	1.51
0.89	1.5238	13.70	19.05	15.89	0.62	1.51
0.90	1.5476	13.76	19.05	15.89	0.61	1.51
0.91	1.5714	13.81	19.05	15.89	0.60	1.51
0.92	1.5952	13.87	19.05	15.89	0.59	1.51
0.93	1.6190	13.93	19.05	15.89	0.58	1.51
0.94	1.6429	13.98	19.05	15.89	0.57	1.51
0.95	1.6667	14.04	19.05	15.89	0.56	1.51
0.96	1.6905	14.09	19.05	15.89	0.55	1.51
0.97	1.7143	14.14	19.05	15.89	0.54	1.51
0.98	1.7381	14.19	19.05	15.89	0.53	1.51
0.99	1.7619	14.24	19.05	15.89	0.52	1.51
1.00	1.7857	14.29	19.05	15.89	0.51	1.51